POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Buffer solutions are important part of the industrial and biochemical processes. Explain with an example why pH of a buffer solution does not change significantly on small addition of acids or bases. 8

   A buffer solution contains 0.2M ammonium hydroxide and 0.2 M ammonium chloride. What will be the pH of resulting solution when 1 ml HCl is added? pKb for NH_4OH is 4.75. Consider total volume of buffer solution is unchanged after adding HCl.

   b) Define Galvanic corrosion? Explain the mechanism of rusting of iron in terms of both chemical and electrochemical corrosion. 7

   What is the concentration of Ni^{2+} in the cell at 27°C, if the emf is 0.601V?

   $Ni(s)/Ni^{2+}$ (?) // Cu^{2+}(0.75M) /Cu(s)

   Standard electrode potential of Nickel and Lead are –0.24V and 0.13V respectively.

2. a) Differentiate between electron affinity and electronegativity? How electronegativity varies on moving across the period and on descending down the group. 3+4

   b) What are transition elements? Explain the following features of transition elements with examples. 4+4

      i. Electronic configuration

      ii. Atomic radii

3. a) Define carbonation. How are carbocation stabilized? Write any two methods of formation of carbocation. 2+2+4

   b) State and illustrate mechanism of Markovnikov's rule. 7

4. a) What are the silicone resins? Explain the types of silicone with their uses. 8

   b) Explain the mechanism of additional polymerization. How natural rubber is different from synthetic rubber? 7

5. a) Write the principle of Chromatography. Write the important applications of thin layer chromatography. 3+5

   b) Define cement. Explain its manufacturing process. 2+5

6. a) Define water pollution. Write causes, effects and control measures of water pollution. 3+5

   b) Discuss about the air pollution caused by oxides of non-metals 3+4

7. Write short notes on: (**Any two**) 2×5

   a) Standard Hydrogen Electrode (SHE)

   b) Thermoplastics and Thermosetting plastics.

   c) Soil pollution